

11th Module

CFG + Context free grammar

→ It is defined by 4 tuples

$$G \rightarrow (V_n, \Sigma, P, S)$$

$V_n \rightarrow$ finite non-empty set of non-terminals.
written by (uppercase letters).

$\Sigma \rightarrow$ finite set of terminals represented by small case letters / digits / operators (*, +, ...).

$P \rightarrow$ Production rules.

$S \rightarrow$ start symbol.

ex. $\rightarrow S \rightarrow aA$ — ①
Start symbol $A \rightarrow aA \mid bA \mid \epsilon$ — ②

Production rule.

$$V_n = \{S, A\}$$

$$\Sigma = \{a, b\}$$

$G \rightarrow$ is a CFG if every production rule is of the form

$$A \rightarrow \alpha$$

$$A \in V_n$$

$$\alpha \in (V_n \cup \Sigma)^*$$

Q. Construct a CFG for a language $L = W^c W^R$ where

$$W \in \{a, b\}^* \quad L = \{aabcba, abbbba, \dots\}$$

Sol.

$$S \rightarrow aSA \quad \text{--- ①}$$

$$S \rightarrow bSB \quad \text{--- ②}$$

$$S \rightarrow \epsilon \quad \text{--- ③}$$

$$a \quad abacaba$$

$$S \rightarrow aSa$$

$$abSa$$

$$abababa$$

$$abacaba$$

Q. Construct a CFH for any language for having any no. of a's.
 $\Sigma = \{a, b\}$.

Sol. $L = \{ \epsilon, a, aa, aaa, \dots \}$.

$$S \rightarrow \epsilon$$

$$S \rightarrow aS.$$

$$S \rightarrow aS$$

$$aaS$$

$$aaaaS$$

$$aaaaaS$$

$$aaaa\epsilon.$$

Q. Construct a CFH for the language $L = a^n b^{2n}$ where $n \geq 1$.

Sol. $L = \{ abb, aabbbb, \dots \}$.

$$S \rightarrow aSbb.$$

$$S \rightarrow abb.$$

Take a production rule of smallest string always.

$\Rightarrow$ Derivation Tree.

• It is a graphical representation of production rule for a given CFH.

• It is called a Parse Tree.

• Parse tree contains the following properties -

- (i) The root node is always a start symbol.
- (ii) The derivation ^{tree} is read from left to right.
- (iii) The leaf nodes are always the terminal nodes.
- (iv) The intermediate nodes are always the non-terminal nodes.

Q. Construct a derivation tree for a given production rule.

Given: $E \rightarrow E + E$

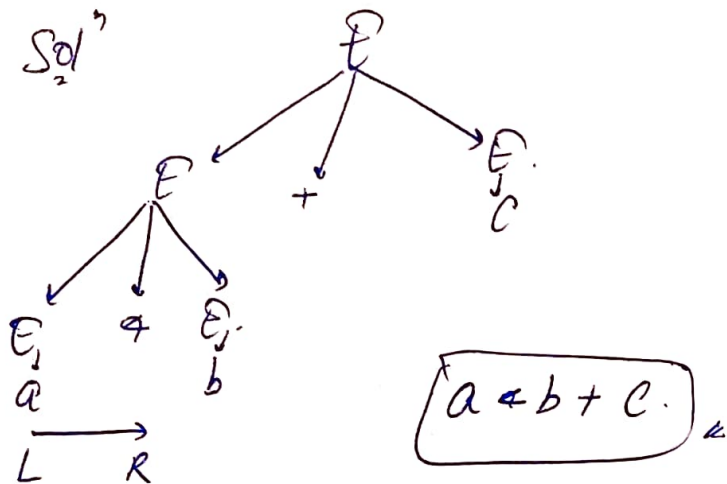
To Proof:

$$\text{string} \rightarrow a + b + c.$$

$$E \rightarrow E + E$$

$$E \rightarrow a/b/c.$$

Solⁿ



Q. Construct the derivation tree for a string aabbabba

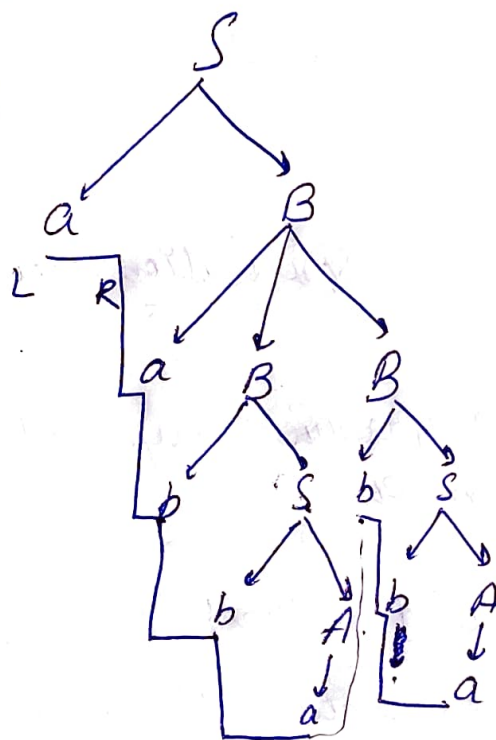
$$S \rightarrow aB / bA$$

$$A \rightarrow a / aS / bAA$$

$$B \rightarrow b / bS / aBB$$

more than one parse tree possible as answers.

aabbabba.



Q. $E \rightarrow E + E$

$$E \rightarrow E * E$$

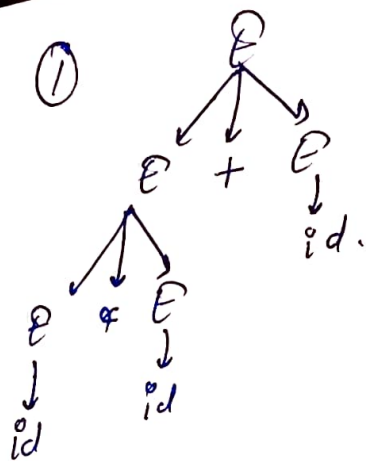
$$E \rightarrow (E)$$

$$E \rightarrow id$$

① $id + id + id$

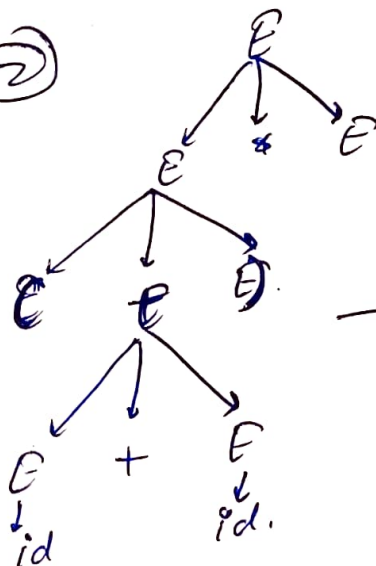
② $(id + id) + id$

①



$$\rightarrow id + id + id$$

②



$$\rightarrow (id + id) * id$$

Q: Construct a derivation tree from LMD & RMD 'abb'.

leftmost Derivation rightmost Derivation

$$S \rightarrow AB \mid \epsilon$$

$$A \rightarrow aB$$

$$B \rightarrow Sb$$

Solⁿ

LMD.

replacing factor eqⁿ

$$S \rightarrow \underline{A}B$$

$$S \rightarrow aB$$

$$S \rightarrow a\underline{B}B$$

$$B \rightarrow Sb$$

$$S \rightarrow aSbB$$

$$S \rightarrow \epsilon$$

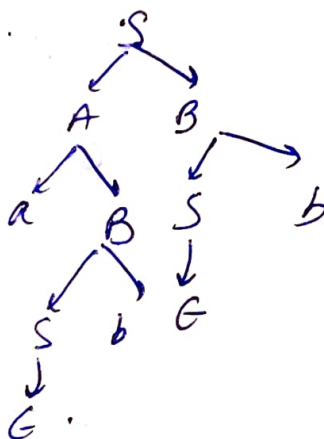
$$S \rightarrow a\epsilon bB$$

$$S \rightarrow ab\underline{B}$$

$$S \rightarrow abSb$$

$$S \rightarrow ab\epsilon b$$

$$S \rightarrow abb$$



RMD

$S \rightarrow A\underline{B}$ $B \rightarrow Sb$

$S \rightarrow A \underline{S} b$ $S \rightarrow \epsilon$

$S \rightarrow A \epsilon b$

$S \rightarrow \underline{A} b$

$S \rightarrow a \underline{B} b$

$S \rightarrow a \underline{S} b b$

$S \rightarrow a \epsilon b b$

$S \rightarrow a b b$

Q: 00101

LMD & RMD.

$S \rightarrow A B$

$A \rightarrow 0 A / \epsilon$

$B \rightarrow 0 B / 1 B / \epsilon$

LMD

$S \rightarrow \underline{A} B$

$S \rightarrow 0 \underline{A} B$

$S \rightarrow 00 \underline{A} B$

$S \rightarrow 00 \epsilon B$

$S \rightarrow 00 1 \underline{B}$

$S \rightarrow 00 10 \underline{B}$

$S \rightarrow 00 10 1 \underline{B}$

$S \rightarrow 00 10 / \epsilon$

$S \rightarrow 00 10 1$

RMD

$S \rightarrow A \underline{B}$

$S \rightarrow A 10 \underline{B}$

$S \rightarrow A 10 1 \underline{B}$

$S \rightarrow A 10 1 \epsilon$

$S \rightarrow 0 \underline{A} 10 1$

$S \rightarrow 00 \underline{A} 10 1$

$S \rightarrow 00 10 1$

⇒

Ambiguity in Grammar

A grammar is said to be ambiguous if it contains more than 1 LMD or more than 1 RMD for a given input string.

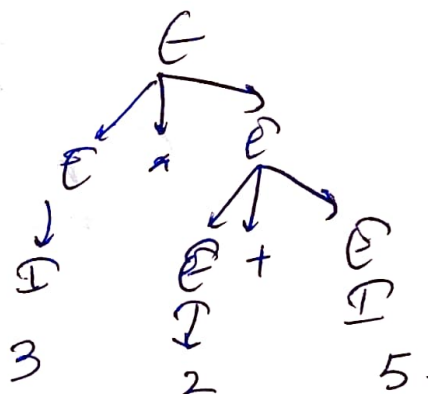
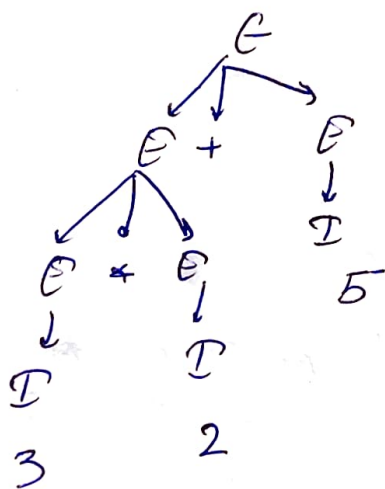
Q₄: Check whether the grammar is ambiguous $3 * 2 + 5$

$$E \rightarrow I$$

$$E \rightarrow E + E$$

$$E \rightarrow E * E$$

$$I \rightarrow \epsilon / 0 / 1 / 2 / 3 \dots 9$$



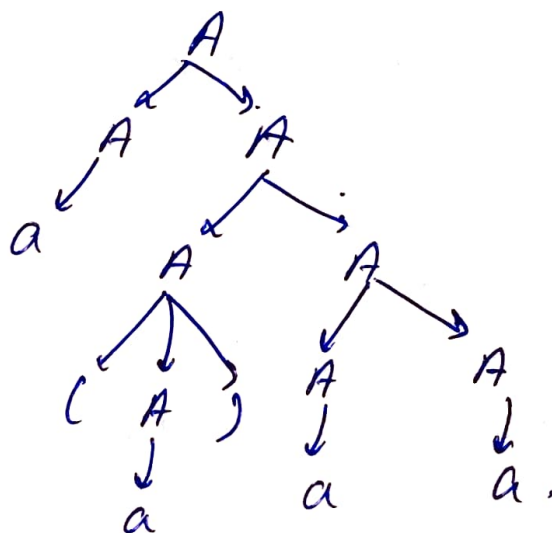
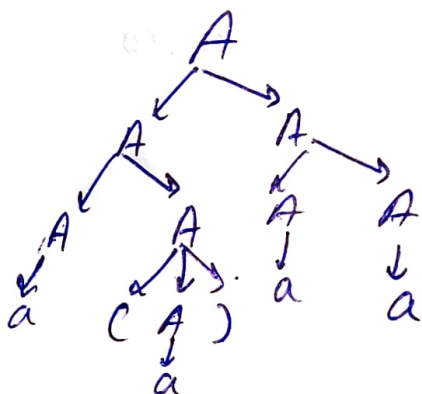
Q₅

$$a(A)aa$$

$$A \rightarrow AA$$

$$A \rightarrow (A)$$

$$A \rightarrow a$$



Q.1 $S \rightarrow S + S / S * S / a / b$
 String $a + a * b$

Q.2. $S \rightarrow sbS / a$
 String $abababa$.

$\Rightarrow$ Simplification of CFLs

- Remove useless symbol $\rightarrow$ production which is not used in deriving grammar.
- Remove null production
$$\left. \begin{array}{l} A \rightarrow \epsilon \\ B \rightarrow \epsilon \end{array} \right\}$$
- Remove unit production
$$\left. \begin{array}{l} A \rightarrow B \\ B \rightarrow C \end{array} \right\}$$

$\rightarrow$ A symbol can be useless if it does not appear on the right hand side of the production and it does not take part in the derivation of the given string.

Ex. $T \rightarrow aaB / (abA) aAT$. String $inabba$.

$A \rightarrow (aa) \checkmark$

$B \rightarrow ab / b$

$C \rightarrow (ad) \checkmark$

equivalent to grammar G

Q. Find a reduced grammar, having production rules

$S \rightarrow AC / B$

$A \rightarrow a$

$C \rightarrow C / BC$

$E \rightarrow aA / e$

solⁿ Phase I
 $T = \{a, c, e\}$
 (Terminal)

$W_1 = \{A, C, E\} \rightarrow$
 (non-Terminal)

$W_2 = \{A, C, E, S\}$

$W_3 = \{A, C, E, S\}$

$W_3 = W_2$

$G' = \{\{A, C, E, S\}, \{a, c, e\}, \{P\}, S\}$

P:
 $S \rightarrow AC$
 $A \rightarrow a$
 $C \rightarrow c$
 $E \rightarrow aA|e$

Phase 2:

$Y_1 = \{S\} \rightarrow$ start symbol

$Y_2 = \{S, A, C\}$

$Y_3 = \{S, A, C, a, c\}$

$Y_4 = \{S, A, C, a, c\}$

$Y_4 = Y_3$

$G'' = \{\{S, A, C\}, \{a, c\}, \{P\}, S\}$

P:
 $S \rightarrow AC$
 $A \rightarrow a$
 $C \rightarrow c$ } reduced grammar.

- Firstly find Terminal
- Secondly find non-Terminal from which the terminal generated.

also check non-Terminal may generate a non-terminal

• Always include left hand side of the production rule.

Q.1 $S \rightarrow S + S / S * S / a / b$
 String $a + a * b$

Q.2. $S \rightarrow sbS / a$
 String ~~ab~~ abababa.

$\Rightarrow$ Simplification of CFLs

- Remove useless symbol $\rightarrow$ production which is not used in deriving grammar.
- Remove null production
$$\begin{array}{l} A \rightarrow \epsilon \\ B \rightarrow \epsilon \end{array} \}$$
- Remove unit production
$$\begin{array}{l} A \rightarrow B \\ B \rightarrow C \end{array} \}$$

$\rightarrow$ A symbol can be useless if it does not appear on the right hand side of the production and it does not take part in the derivation of the given string.

Ex. $T \rightarrow aaB / \textcircled{abA} / aAT$. String $aabba$.

$A \rightarrow \textcircled{aA} \checkmark$

$B \rightarrow ab / b$

$C \rightarrow \textcircled{bd} \checkmark$

equivalent to grammar G

Q. Find a reduced grammar, having production rules

$S \rightarrow AC / B$

$A \rightarrow a$

$C \rightarrow C / BC$

$E \rightarrow aA / e$

solⁿ Phase I
 $T = \{a, c, e\}$
 (Terminal)

$W_1 = \{A, C, E\} \rightarrow$
 (non-Terminal)

$W_2 = \{A, C, E, S\}$

$W_3 = \{A, C, E, S\}$

$W_3 = W_2$

$G' = \{\{A, C, E, S\}, \{a, c, e\}, \{P\}, S\}$

$P:$
 $S \rightarrow AC$
 $A \rightarrow a$
 $C \rightarrow c$
 $E \rightarrow aA/c$

Phase 2:

$\gamma_1 = \{S\} \rightarrow$ start symbol.

$\gamma_2 = \{S, A, C\}$

$\gamma_3 = \{S, A, C, a, c\}$

$\gamma_4 = \{S, A, C, a, c\}$

$\gamma_4 = \gamma_3$

$G'' = \{\{S, A, C\}, \{a, c\}, \{P\}, S\}$

$P:$
 $S \rightarrow AC$
 $A \rightarrow a$
 $C \rightarrow c$ } reduced grammar.

- Firstly find Terminal
- Secondly find non-Terminal from which the terminal generated.

also check non-Terminal may generate a non-terminal

... Always include left hand side of the production rule.

Q: find the reduced grammar equivalent to grammar G having production rules

$$S \rightarrow AB/CA$$

$$B \rightarrow B/AB$$

$$A \rightarrow a$$

$$C \rightarrow aB/b$$

solⁿ phase 1:

$$T = \{a, b\}$$

$$W_1 = \{A, C\}$$

$$W_2 = \{A, C, S\}$$

$$W_3 = \{A, C, S\}$$

$$W_3, W_2$$

$$G' = \{ \{A, C, S\}, \{a, b\}, \{P, S\} \}$$

$$P: S \rightarrow CA$$

$$A \rightarrow a$$

$$C \rightarrow b$$

phase 2:

$$Y_1 = \{S\}$$

$$Y_2 = \{S, C, A\}$$

$$Y_3 = \{S, C, A, a, b\}$$

$$Y_4 = \{S, C, A, a, b\}$$

$$G'' = \{ \{S, C, A\}, \{a, b\}, \{P\}, \{S\} \}$$

$$P: S \rightarrow CA$$

$$A \rightarrow a$$

$$C \rightarrow b$$

} reduced
gr

Q. $S \rightarrow aAa$
 $A \rightarrow sb/bCC/aAa$
 $C \rightarrow abb/DD$
 $E \rightarrow aC$
 $D \rightarrow aDA$

Phase I $\rightarrow$ जिससे derive ho
 raha hai (LHS)
 Phase II $\rightarrow$ जो derive ho raha
 hai. (RHS).

$$T = \{a, b\}$$

$$W_1 = \{C\}$$

$$W_2 = \{C, A, \epsilon\}$$

$$W_3 = \{C, A, S, \epsilon\}$$

$$W_4 = \{C, A, S, E\} \quad W_5 = \{C, A, S, \epsilon\}$$

$$W_5 = W_4$$

$$G'' = \{\{C, A, S, E\}, \{a, b\}, \{P\}, \{S\}\}$$

P:
 $S \rightarrow aAa$
 $A \rightarrow sb/bCC$
 $C \rightarrow abb$
 $E \rightarrow aC$

Phase II:

$$Y_1 = \{S\}$$

$$Y_2 = \{S, A\}$$

$$Y_3 = \{S, A, a, \epsilon\}$$

$$Y_4 = \{S, A, C, a, b\}$$

$$Y_5 = \{S, A, C, a, b\}$$

$$Y_4 = Y_5$$

$$G'' = \{\{S, A, C\}, \{a, b\}, \{P\}, \{S\}\}$$

P:
 $S \rightarrow aAa$
 $A \rightarrow sb/bCC$
 $C \rightarrow abb$

Removal of Null production

$$\begin{array}{l} A \rightarrow \epsilon \\ B \rightarrow \epsilon \end{array} \left. \vphantom{\begin{array}{l} A \rightarrow \epsilon \\ B \rightarrow \epsilon \end{array}} \right\} \text{Null production.}$$

Q: Remove all the null production from the given grammar.

Firstly take where
we get the null

$$S \rightarrow aS/AB$$

$$A \rightarrow \epsilon/a$$

then put the
null non-terminal
anywhere we
get the null

$$B \rightarrow \epsilon/b$$

$$D \rightarrow b/aAbB$$

$$W_1 = \{A, B\}$$

$$W_2 = \{A, B, S\}$$

$$W_3 = \{A, B, S\}$$

$$W_3 \equiv W_2$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$D \rightarrow b$$

Put one by one for
every character.

$$S \rightarrow aS/AB/a|B|A$$

$$D \rightarrow aAbB/abB/aAb/\epsilon b$$

$$Q: S \rightarrow aA$$

$$A \rightarrow b/\epsilon$$

$$W_1 = \{A\}$$

$$W_2 = \{A\}$$

$$S \rightarrow aA/a \quad W_2 = W_1$$

$$A \rightarrow b$$

$$Q: S \rightarrow ABAC/\cancel{AAC}/\cancel{ABA}/\cancel{AA}$$

$$A \rightarrow BC/\cancel{C}/\cancel{B}$$

$$B \rightarrow b/\epsilon$$

$$C \rightarrow D/\epsilon$$

$$D \rightarrow d$$

$$W_1 = \{B, C\}$$

$$W_2 = \{B, C, A\}$$

$$W_3 = \{B, C, A\}$$

$$W_3 = W_2$$

$$S \rightarrow ABaC | BaC | AaC | ABA | aC | Aa | Ba | a.$$

$$A \rightarrow Bc | C | B.$$

$$B \rightarrow b$$

$$C \rightarrow D$$

$$D \rightarrow d$$

Removal of unit production

$$S \rightarrow A | bb.$$

$$A \rightarrow B | b$$

$$B \rightarrow S | a.$$

Always remove non-terminal

$$W_1(S) = \{S\}$$

$$W_2(S) = \{S, A\}$$

$$W_3(S) = \{S, A, B\}$$

$$W_4(S) = \{S, A, B\}$$

$$[W_4(S) = W_3(S)]$$

$$W_1(A) = \{A\}$$

$$W_2(A) = \{A, B\}$$

$$W_3(A) = \{A, B, S\}$$

$$W_4(A) = \{A, B, S\}$$

$$[W_4(A) = W_3(A)]$$

$$W_1(B) = \{B\}$$

$$W_2(B) = \{B, S\}$$

$$W_3(B) = \{B, S, A\}$$

$$W_4(B) = \{B, S, A\}$$

$$[W_4(B) = W_3(B)]$$

4.

use transitive property

$$S \rightarrow b | a | bb$$

$$A \rightarrow a | b | bb$$

$$B \rightarrow bb | b | a.$$

$$\begin{aligned} S &\rightarrow 0A/1B/C \\ A &\rightarrow 0S/00 \\ B &\rightarrow 1/A \\ C &\rightarrow 0/1 \end{aligned}$$

$$\begin{aligned} W_1(S) &= \{S\} \\ W_2(S) &= \{S, C\} \\ W_3(S) &= \{S, C\} \end{aligned}$$

$$W_1(A) = \{A\}$$

$$W_2(A) = \{A\}$$

$$W_2(A) = W_1(A)$$

$$W_1(C) = \{C\}$$

$$W_2(C) = \{C\}$$

$$W_2(C) = W_1(C)$$

$$W_3(S) = W_2(S)$$

$$W_1(B) = \{B\}$$

$$W_2(B) = \{B, A\}$$

$$W_3(B) = \{B, A\}$$

$$W_3(B) = W_2(B)$$

$$S \rightarrow 0A/1B/01$$

$$A \rightarrow 0S/00$$

$$B \rightarrow 1/0S/00$$

$$C \rightarrow 01$$

Normal of CFG

① Chomsky's Normal form (CNF)

A CFG is in CNF if all the production rules are one of the following form.

- A start symbol generating null. ($S \rightarrow \epsilon$).
- A non-terminal generating two non-terminals. ($A \rightarrow BC$).
- A non-terminal generating terminal ($A \rightarrow a$).

$$G_1: \begin{aligned} S &\rightarrow AB \\ S &\rightarrow C \\ A &\rightarrow a \\ B &\rightarrow b. \end{aligned}$$

$$G_2: \begin{aligned} S &\rightarrow oA \\ A &\rightarrow a \\ B &\rightarrow c. \end{aligned}$$

(NOT in CNF form).

(all are in CNF form).

Convert the following CFG into CNF form.

$$S \rightarrow aSa / bSb / aa / bb / a / b.$$

Step 1: Remove all the null and unit production from the grammar.

when non-terminal generate non-terminal.

Step 2: Add productions already in CNF.

$$S \rightarrow a / b.$$

Step 3:

$$S \rightarrow aSa \quad \text{let } N_a \rightarrow a$$

$$S \rightarrow N_a S N_a$$

$$\begin{aligned} S &\rightarrow N_a S_1 \\ S_1 &\rightarrow S N_a \end{aligned}$$

$$\begin{aligned} S &\rightarrow N_a N_a \\ S &\rightarrow N_b N_b \end{aligned}$$

$$S \rightarrow bSb \quad \text{let } N_b \rightarrow b$$

$$S \rightarrow N_b S N_b$$

$$\begin{aligned} S &\rightarrow N_b S_2 \\ S_2 &\rightarrow S N_b \end{aligned}$$

ALL CNF forms are in box.
write all here.

$$S \rightarrow aXbX / aXb / aXb / ab / \epsilon.$$

$$X \rightarrow aY / bY / a / b$$

$$Y \rightarrow aY / bY / a / b / c.$$

Start symbol generates null we can remove it..

Solⁿ

Step II $\rightarrow S \rightarrow aXbX$

let. $N_a \rightarrow a$
 $N_b \rightarrow b$

$S \rightarrow NaXNbX$

$S \rightarrow NaS_1$
 $S_1 \rightarrow NaS_2$
 $S_2 \rightarrow NbX$

$4 \rightarrow 3$
 $3 \rightarrow 2$

$S \rightarrow abX$

let. $N_a \rightarrow a$
 $N_b \rightarrow b$

$S \rightarrow NaNbX$

$S \rightarrow NaS_4$

$S_4 \rightarrow NbX$

$S \rightarrow aXb$

$S \rightarrow NaXNb$

$S \rightarrow NaS_3$

$S_3 \rightarrow XNb$

$S \rightarrow ab$

$S \rightarrow NaNb$

$X \rightarrow aY$

$X \rightarrow bY$

$X \rightarrow NaY$

$X \rightarrow NbY$

$Y \rightarrow aY$

$Y \rightarrow bY$

$Y \rightarrow NaY$

$Y \rightarrow NbY$

All productions

$S \rightarrow NaS_1$

$S \rightarrow NaS_1$

$S_1 \rightarrow NaS_2$

$S_2 \rightarrow XNb$

$S_2 \rightarrow NbX$

$S \rightarrow NaNb$

$S \rightarrow NaS_1$

$X \rightarrow NaY$

$S_4 \rightarrow NbX$

$X \rightarrow NbY$

$X \rightarrow a$

$Y \rightarrow NaY$

$X \rightarrow b$

$Y \rightarrow a$

$Y \rightarrow NbY$

$Y \rightarrow b$

$Na \rightarrow a$

$Y \rightarrow c$

$Nb \rightarrow b$

Q.

$$S \rightarrow bA / aB$$

$$A \rightarrow bAA / aS / a$$

$$B \rightarrow aBB / b / a$$

Solⁿ
 Step II $\rightarrow S \rightarrow aXbX$
 let. $N_a \rightarrow a$
 $N_b \rightarrow b$
 $S \rightarrow NaXNbX$

$S \rightarrow NaS_1$
 $S_1 \rightarrow NaS_2$
 $S_2 \rightarrow NbX$

$4 \rightarrow 3$
 $3 \rightarrow 2$

$S \rightarrow abX$

let. $N_a \rightarrow a$
 $N_b \rightarrow b$

$S \rightarrow NaNbX$

$S \rightarrow NaS_4$
 $S_4 \rightarrow NbX$

$S \rightarrow aXb$

$S \rightarrow NaXNb$

$S \rightarrow NaS_3$
 $S_3 \rightarrow XNb$

$S \rightarrow ab$

$S \rightarrow NaNb$

$X \rightarrow aY$

$X \rightarrow bY$

$X \rightarrow NaY$

$X \rightarrow NbY$

$Y \rightarrow aY$

$Y \rightarrow bY$

$Y \rightarrow NaY$

$Y \rightarrow NbY$

All productions

$S \rightarrow NaS_1$	$S \rightarrow NaS_1$
$S_1 \rightarrow NaS_2$	$S_2 \rightarrow XNb$
$S_2 \rightarrow NbX$	$S \rightarrow NaNb$
$S \rightarrow NaS_1$	$X \rightarrow NaY$
$S_4 \rightarrow NbX$	$X \rightarrow NbY$
$X \rightarrow a$	$Y \rightarrow NaY$
$X \rightarrow b$	$Y \rightarrow NbY$
$Y \rightarrow a$	
$Y \rightarrow b$	$Na \rightarrow a$
$Y \rightarrow c$	$Nb \rightarrow b$

Q. 4

$$S \rightarrow bA / aB$$

$$A \rightarrow bAA / aS / a$$

$$B \rightarrow aBB / b / a$$

VIP 5 marks.

Greibach normal form

A CFG is in GNF if all the production rules satisfy the following condition:

① start symbol can generate null.

$$S \rightarrow \epsilon.$$

② A non-terminal can generate terminal.

$$A \rightarrow a.$$

③ A non-terminal generating a terminal which is followed by any number of non-terminals.

$$A \rightarrow aBCD.$$

∴

$$G_1: S \rightarrow aAB/aB$$

$$A \rightarrow aA/a$$

$$B \rightarrow bB/b$$

(in GNF form)

$$G_2: S \rightarrow aAB/aB$$

$$A \rightarrow aA/\epsilon$$

$$B \rightarrow bB/\epsilon.$$

(Not in GNF form).

Step to convert CNF to GNF:

Step 1: Remove all unit and null productions.

Step 2: Check whether the CFG is already in CNF. if not then convert it into CNF.

Step 3: Change the names of the non-terminals symbols into some of A_i in ascending order of i .

Q. $S \rightarrow CA|BB$
 $B \rightarrow b|SB$
 $C \rightarrow b$
 $A \rightarrow a$.

Sol. Step 1: NO null & unit production.

Step 2: Given grammar is in CNF form.

Step 3: Replace S with A_1 .

C with A_2
 A with A_3
 B with A_4 .

$$A_1 \rightarrow A_2 A_3 / A_4 A_4$$

$$A_4 \rightarrow b / A_1 A_4$$

$$A_2 \rightarrow b$$

$$A_3 \rightarrow a$$

Take in sequence as we get the non-terminal like in

$$S \rightarrow CA|BB$$

$$A_1 \rightarrow A_2 A_3 A_4$$

such that

Step 4: After the rules the non-terminal are in ascending order if the production is of the form.
 $A_i \rightarrow A_j X$ then

if $i < j$ then production is correct

if $i \geq j$ then we have to resolve the production.

$$A_4 \rightarrow b / A_1 A_4$$

put A_1 in A_4 .

$$A_4 \rightarrow b / A_2 A_3 A_4 / A_4 A_4 A_4$$

put A_2 in A_4 .

$$A_4 \rightarrow b / b A_3 A_4 / A_4 A_4 A_4$$

$$A_4 \rightarrow A_4 A_4 A_4$$

(when we get same value on both side of A_i then it's called left recursion).
 Ok.

Step 5: Remove left recursion introduce a new variable (Z).

$$Z \rightarrow A_4 A_4 Z / A_4 A_4$$

One A_4 is removed.

$$A_4 \rightarrow b / b Z / b A_3 A_4 / b A_3 A_4 Z$$

$$A_2 \rightarrow b$$

$$A_3 \rightarrow a$$

replace A_2 in A_1 and A_4 in A_1 .

$$A_1 \rightarrow b A_3 / b A_4 / b Z A_4 / b A_3 A_4 A_4 / b A_3 A_4 Z A_4$$

put A_4 in Z .

$$Z \rightarrow b A_4 Z / b Z A_4 Z / b A_3 A_4 A_4 Z / b A_3 A_4 Z A_4 Z / b A_4 / b Z A_4 / b A_3 A_4 A_4 / b A_3 A_4 Z A_4$$

5/11/24

Q: Construct a grammar in GNF form equivalent to grammar:

$$S \rightarrow AA/a$$

$$A \rightarrow SS/b$$

Sol:

replace S with A

" A with A_2

$$A_1 \rightarrow A_2 A_2 / a \quad \text{--- (1)}$$

$$A_2 \rightarrow A_1 A_1 / b \quad \text{--- (2)}$$

put (2) in (1)

$$A_2 \rightarrow A_2 A_2 A_1 / a A_1 / b$$

Introduce a new variable Z .

$$Z \rightarrow A_2 A_1 / A_2 A_2$$

$$A_2 \rightarrow a A_1 / a A_1 Z / b / b Z$$

$$A \rightarrow a A A_2 / a A_1 z A_1 z / b A_2 / b z A_2$$

$$z \rightarrow a A A_1 / a A_1 z A_1 / b A_1 / b z A_1 / a A A_1 z / a A_1 z A_1 z / b A_1 z / b z A_1 z$$

Q:

$$\begin{aligned} S &\rightarrow AB \\ A &\rightarrow BS / b \\ B &\rightarrow SA / a \end{aligned}$$

Conversion from CFG to PDA :

Rules

- ① $R_1: \delta(q, \epsilon, A) = \{(q, \alpha) \mid A \rightarrow \alpha \text{ is in Production } P\}$ start symbol
- ② $R_2: \delta(q, a, a) = \{(q, \epsilon)\}$ for every a in Σ .

Note: There is no pushing operation while converting to CFG to PDA, only pop operation is used.

Q. Construct a PDA which is equivalent to the CFG.

$$S \rightarrow 0BB$$

$$B \rightarrow 0S \mid 1S \mid 0 \quad \text{Test whether } 010000 \text{ is in } N(A)$$

$\hookrightarrow$ PDA name.

Sol. Define PDA:

$$A = \{q, \Sigma, \Gamma, \delta, q_0, z_0, F\}.$$

$$= \{ \overset{q}{\{q\}}, \overset{\Sigma}{\{0, 1\}}, \overset{\Gamma}{\{S, B, 0, 1\}}, \delta, q_0, z_0, F \}.$$

$$R_1: \delta(q, \epsilon, S) = (q, 0BB)$$

$A \rightarrow \alpha$
P.

$$R_2: \delta(q, \epsilon, B) = \{(q, 0S), (q, 1S), (q, 0)\}.$$

$$R_3: \delta(q, 0, 0) = (q, \epsilon)$$

$$R_4: \delta(q, 1, 1) = (q, \epsilon).$$

$$(q, 010000, S)$$

$$\vdash (q, \emptyset 10000, \emptyset BB)$$

$$\vdash (q, \text{10000}, \text{15B})$$

$$\vdash (q, \text{0000}, \text{0BBB})$$

$$\vdash (q, \text{0000}, \text{0000})$$

$$\vdash (q, \epsilon, \epsilon)$$

so the given ^{string} ~~grammar~~ is accepted by the grammar.

$$Q. S \rightarrow asa / bsb / c.$$

abbcbbba is in $N(CA)$.

ww^r.

solⁿ Define PDA:

$$A = \{q, \Sigma, \Gamma, \delta, q_0, z_0, F\}$$

$$= \{ \{q\}, \{a, b, c\}, \{S, a, b, c\}, \delta, q_0, z_0, F \}$$

$$R_1 : \delta(q_0, \epsilon, S) = \{(q_0, asa), (q_0, bsb), (q_0, c)\}$$

$$(q_0, a, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, b) = (q_2, \epsilon)$$

$$\vdash (q, abbcbbba, asa) \quad \delta(q_2, c, c) = (q_3, \epsilon)$$

$$\vdash (q, bbcbbba, bsb)$$

$$\vdash (q, cbbba, c)$$

$$\vdash (q, \epsilon, \epsilon)$$

$$\vdash (q, \epsilon, \epsilon)$$

Q.

No need to change the state.

Conversion from PDA to CF VIM Study

Rule

① Construction of set of non-terminals

$$V = \{S\} \cup \{[q, z, q'] \mid q, q' \in Q, z \in \Gamma\}$$

② (i) $S \rightarrow$ Production

$$S \rightarrow [q_0, z_0, q], q \in Q.$$

(ii) for pop operation

$$\delta(q, a, z) \rightarrow (q', \epsilon)$$

$$[q, z, q'] \rightarrow a$$

To find no. of production
No of states $\times$ No of
Stack symbol.

(iii) Push and SKIP.

$$\delta(q, a, z) \rightarrow (q_1, z, z_1, \dots, z_m)$$

$$[q, z, q'] \rightarrow a [a_1, z_1, q_2] [a_2, z_2, q_3] [a_3, z_3, q_4] \dots [a_m, z_m, q']$$

$$q' \in Q$$

$$\text{where } q_1, q_2, q_3, \dots, q_m \in Q.$$

Q: N(A)

$$A = (\{q_0, q_1\}, \{a, b\}, \{z_0, z\}, \delta, q_0, z_0, \emptyset).$$

and δ is given by.

$$\begin{aligned} \delta(q_0, b, z_0) &= (q_0, z z_0) \\ \delta(q_0, \epsilon, z_0) &= (q_0, \epsilon) \\ \delta(q_0, b, z) &= (q_0, z z) \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{aligned} \delta(q_0, a, z) &= (q_1, z) \\ \delta(q_1, b, z) &= (q_1, \epsilon) \\ \delta(q_1, a, z_0) &= (q_0, z_0) \end{aligned}$$

Sol: $G = (V_n, (a, b), P, S)$.

① V_n consist of S $[q_0, z_0, q_0]$ $[q_0, z_0, q_1]$
 $[q_1, z_0, q_0]$ $[q_1, z_0, q_1]$
 $[q_0, z, q_0]$ $[q_0, z, q_1]$.
 $[q_1, z, q_0]$ $[q_1, z, q_1]$.

② S-production

$P_1: S \rightarrow [q_0, z_0, q_0]$
 $P_2: [q_0, z_0, q_1]$.

Push के लिए
4 तरीके

③ $S(q_0, b, z_0) = (q_0, z z_0)$ push.

$P_3: [q_0, z_0, q_0] \rightarrow b [q_0, z, q_0] [q_0, z_0, q_0]$

$P_4: [q_0, z_0, q_0] \rightarrow b [q_0, z, q_1] [q_1, z_0, q_0]$

$P_5: [q_0, z_0, q_1] \rightarrow b [q_0, z, q_0] [q_0, z_0, q_1]$

$P_6: [q_0, z_0, q_1] \rightarrow b [q_0, z, q_1] [q_1, z_0, q_1]$.

④ $S(q_0, \epsilon, z_0) = (q_0, \epsilon)$ pop

$P_7: [q_0, z_0, q_0] \rightarrow \epsilon$

⑤ $S(q_0, b, z) = (q_0, z z)$ push.

$$P_8: [q_0, z, a_0] \rightarrow b [q_0, z, a_0]$$

$$P_9: [q_0, z, a_0] \rightarrow b [q_0, z, a_1] [q_1, z, a_0]$$

$$P_{10}: [q_0, z, a_1] \rightarrow b [q_0, z, a_0] [q_0, z, a_1]$$

$$P_{11}: [q_0, z, a_1] \rightarrow b [q_0, z, a_1] [q_1, z, a_1]$$

$$\textcircled{7} \delta(q_0, a_1, z) = (q_1, z)$$

skip कर लियो 2 & 3 वीं

पुनर्नी state
3 वीं production

$$P_{12}: [q_0, z, a_0] \rightarrow a [q_1, z, a_0]$$

$$P_{13}: [q_0, z, a_1] \rightarrow a [q_1, z, a_1]$$

$$\textcircled{8} \delta(q_1, b, z) = (q_1, e) \quad \textcircled{9} \delta(q_1, a, z_0) = (q_0, z_0)$$

$$P_{14}: [q_1, z, a_1] \rightarrow b$$

$$P_{15}: [q_1, a, a_0] \rightarrow a [q_0, z_0, a_0]$$

$$P_{16}: [q_1, z_0, a_1] \rightarrow a [q_0, z_0, a_1]$$

13/11/24

Turing Machine

→ Turing Machine has infinite size tape.

• Turing Machine is used to accept the recursive enumerable language.

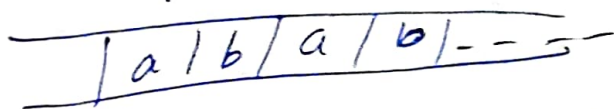
• It can move in both direction and it ~~can~~ not accept the null.

• If the string is not accepted by the language then the machine will halt in non-final state.

• If the string is accepted by the language then the machine will halt in final state.

• Turing machine is a mathematical model which consist an infinite ^{length} model tape divided into cell on which input is given.

Input tape.

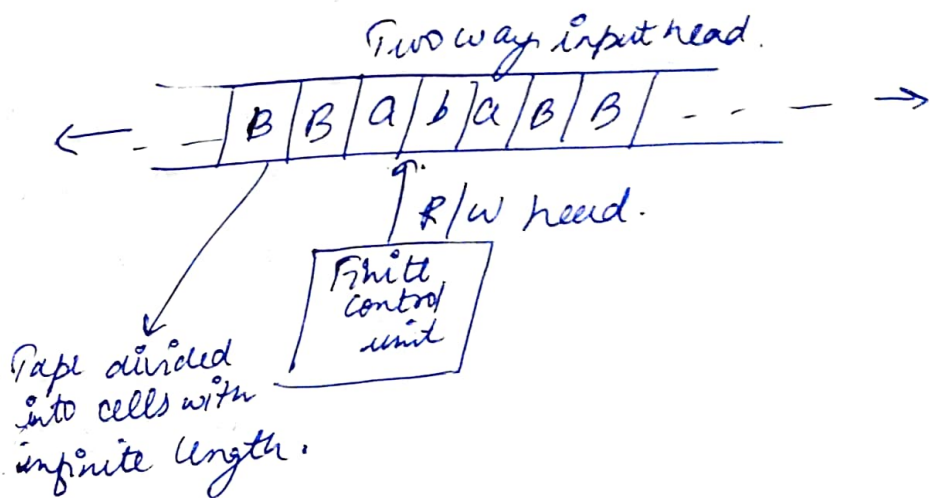


• It consist head which reads the input tape.

After reading an input symbol, it is replaced with another symbol and it moves from one cell to another from right to left.

If the Turing machine reaches to the final state then the ~~input~~ ^{input} string is accepted otherwise rejected.

$B \rightarrow \text{Blank}$



Turing machine consist seven tuples. (C)

C, O, E, S, q_0, F, P, B

$\rho \rightarrow$ Tape alphabet.

$B \rightarrow$ Blank symbol.

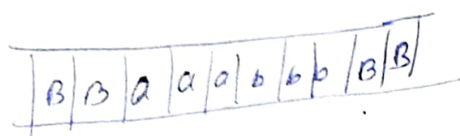
$$\delta : \mathcal{O}_X \rightarrow \mathcal{O}_X$$
$$\mathbb{Q} \times \mathbb{P} \times (L/R)$$

Q: Design a Turing Machine for the language.

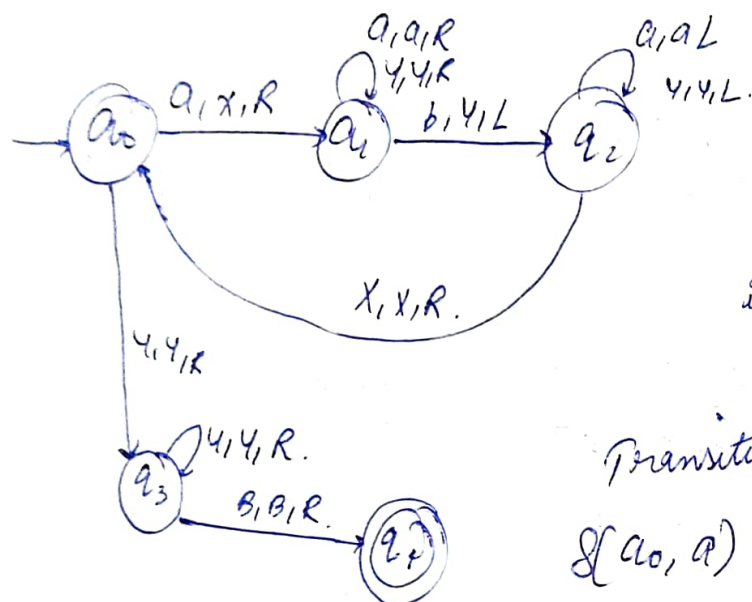
$$L = \{a^n b^n \mid n \geq 1\}.$$

each symbol
replace simulta-
neously.

[aaabbb]



$XaaYbb \quad a \rightarrow X$
 $XXaYb \quad b \rightarrow Y/Y.$
 $XXXYYY.$



a, x, R
 $\downarrow$ replace input from a more right.

Transition Id.

$$\delta(q_0, a) = (q_1, x, R)$$

$$\delta(q_0, y) = (q_3, y, R)$$

$$\delta(q_1, a) = (q_1, a, R)$$

$$\delta(q_1, y) = (q_1, y, R)$$

$$\delta(q_1, b) = (q_2, y, L)$$

$$\delta(q_2, a) = (q_2, a, L)$$

$$\delta(q_2, y) = (q_2, y, L)$$

$$\delta(q_2, x) = (q_0, x, R)$$

$$\delta(q_3, y) = (q_3, y, R)$$

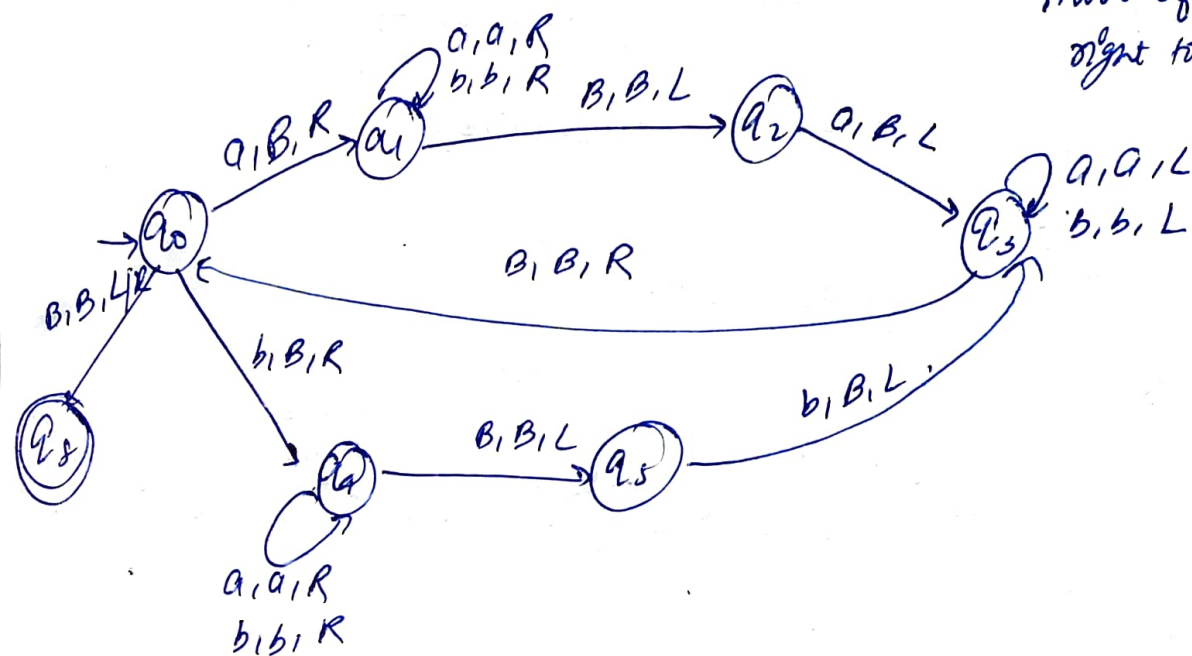
$$\delta(q_3, b) = (q_4, b, R)$$

Q: 2. $\{a^n b^n c^n \mid n \geq 1\}$

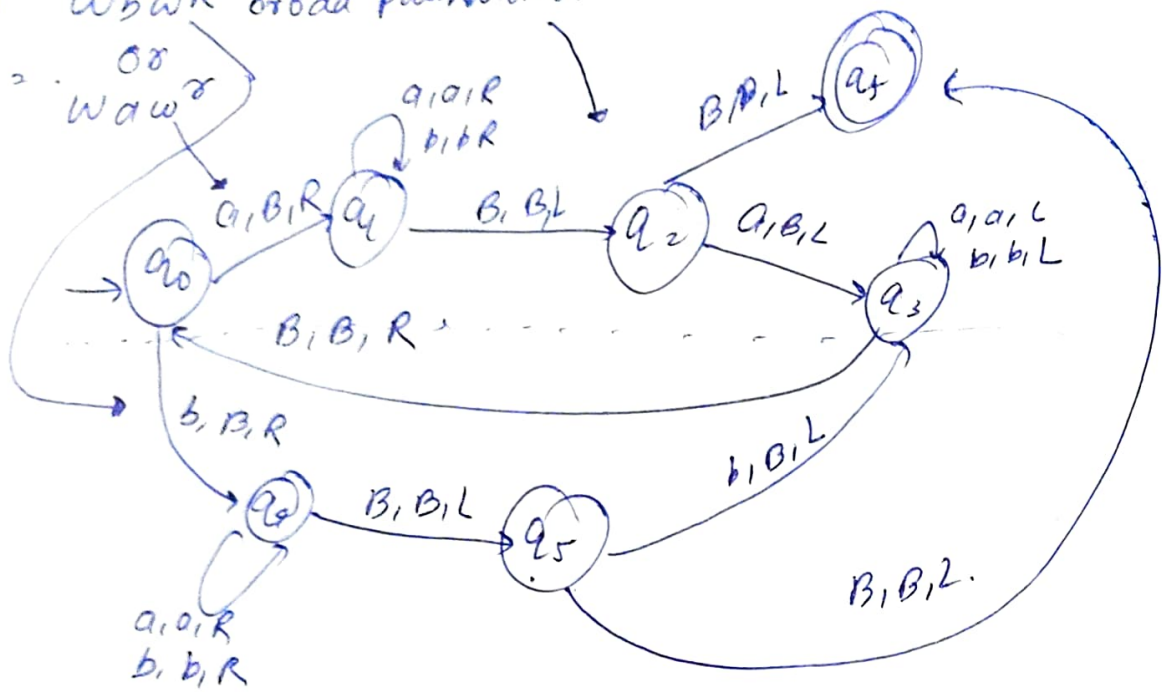
Q: $L = \{ww^R\}$ or even palindrome
 $w \in \{a, b\}^+$

BB aa bb ba ba BB
 $\quad \quad \quad w \quad \quad w^R$

move left to right
 right to left.



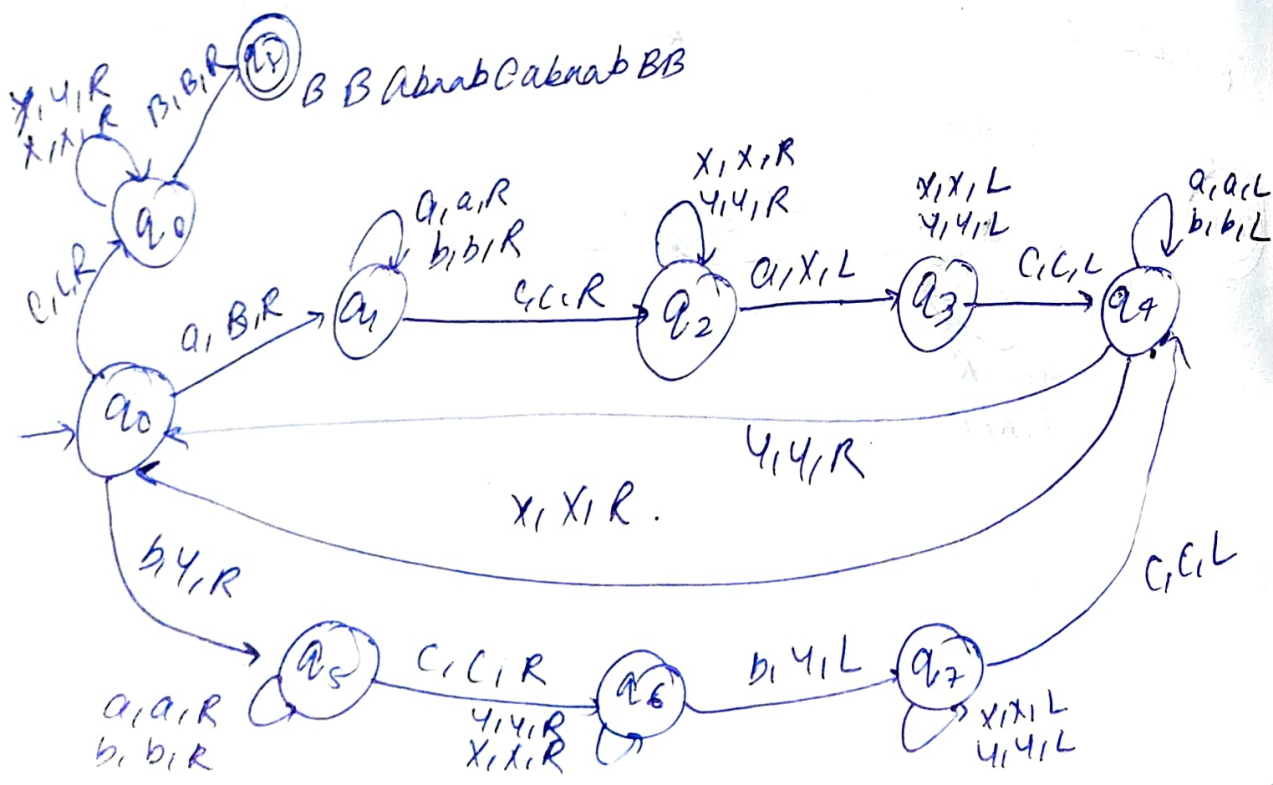
Q: Wbwr or odd palindrome.
 L = or wawr



Odd palindrome

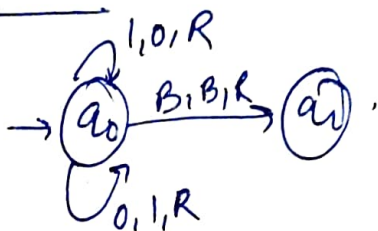
Q: Design a Turing Machine for a given language.

L = $W^c W$ where $W \in \{a, b\}^*$



Q. 2's Complement PM.

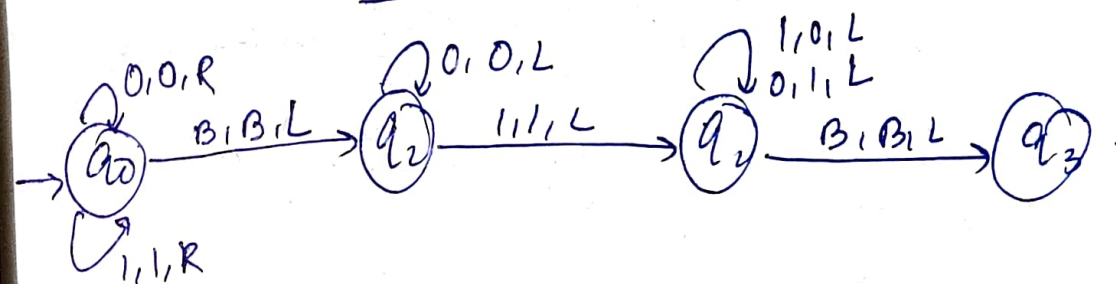
B/B | 1/0 | B/B
0 1



Q. 2's Complement PM.

MSD LSD
B/B | 1/0 | 1/0 | 0/0 | B/B
1 0 0

0 1 1 | 0 0

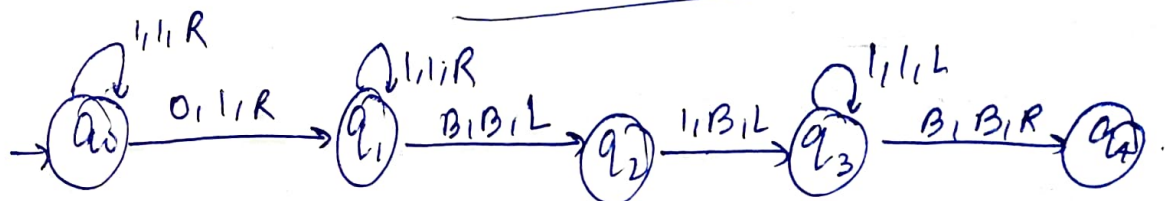


Q. Design PM for addition of two no.

a = 3
b = 4

a + b = 7.

a 1 b
B/B | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | B/B



Q. Design PM of two no's subtraction.

a = 4
b = 3

f(min) = m - n if m > n
0 if m ≤ n

Church Turing thesis :

A functional or natural no.s is computable by an algotm if and only if it is computable by Turing machine.

TM concept has been given by Alan Turing in the year of 1936.

- Anything that can be solved by current digital computer and also be solved by TM.
- Many mathematical model are suggested but no one of them is more powerful than TM. Or we can say that TM is an abstract model for digital computers.

Variations of TM :

→ Different variation of TM are :-

① multi^otape TM

② Non-Deterministic TM

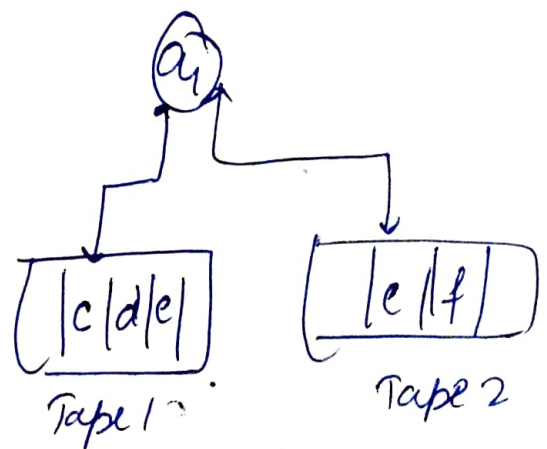
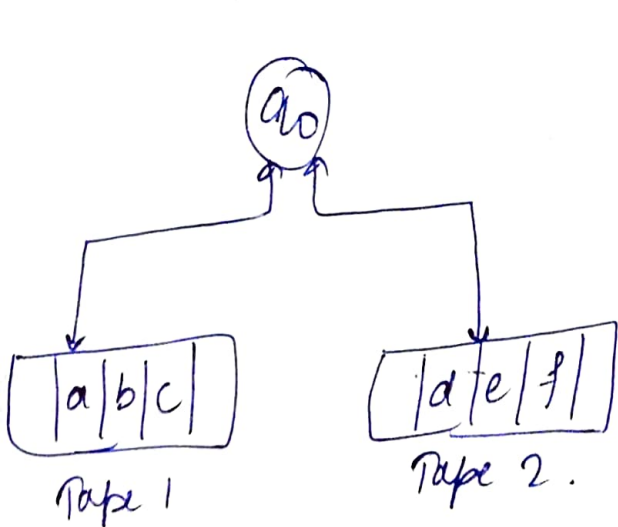
③ multi^ohead TM

④ Offline TM.

⑤ Multi^odimensional TM.

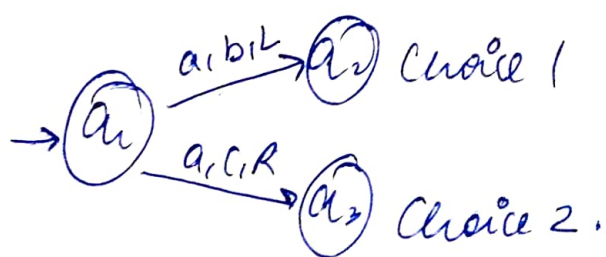
① Multi-tape TM:

- A TM with many tapes.
- Every tapes have their controlled own read and write head.
- $m = \{0, \leq, \delta, q_0, F, B, \gamma\}$.



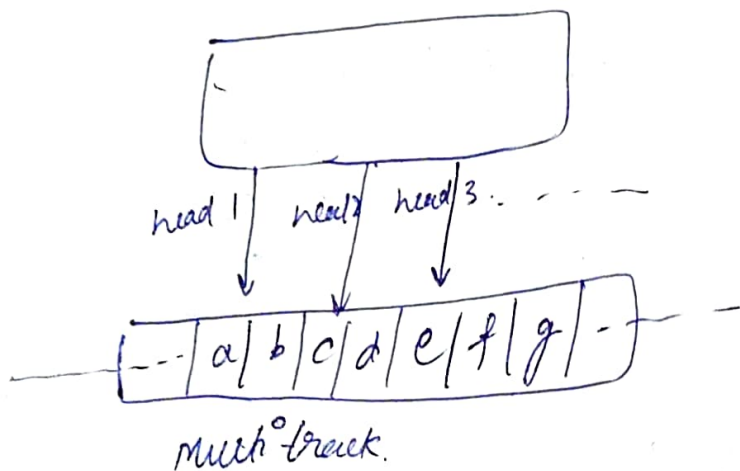
② Non-Deterministic TM:

- It is similar to deterministic TM except that for any input symbol and current state it has a no. of choice.
- A string is accepted by a non-deterministic TM if there is a sequence of moves that leads to a final state.
- Transition function N-D TM is $\delta: Q \times \Sigma \rightarrow 2^{Q \times \Sigma \times \{L, R\}}$.



③ Mult^ohead TM:

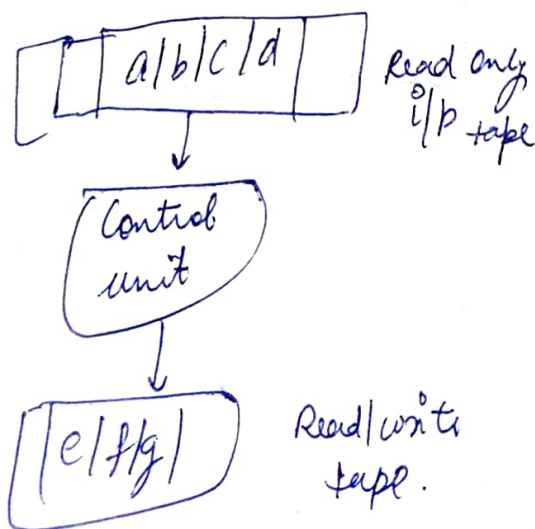
- It has no. of heads instead of one. → each head have independently read & write symbols and move left and right.



	B	1	B	B	B	B	-	-
	1	B	B	B	B	B	-	-
	B	B	B	1	B	B	-	-
	a	b	c	d	e	f	g	

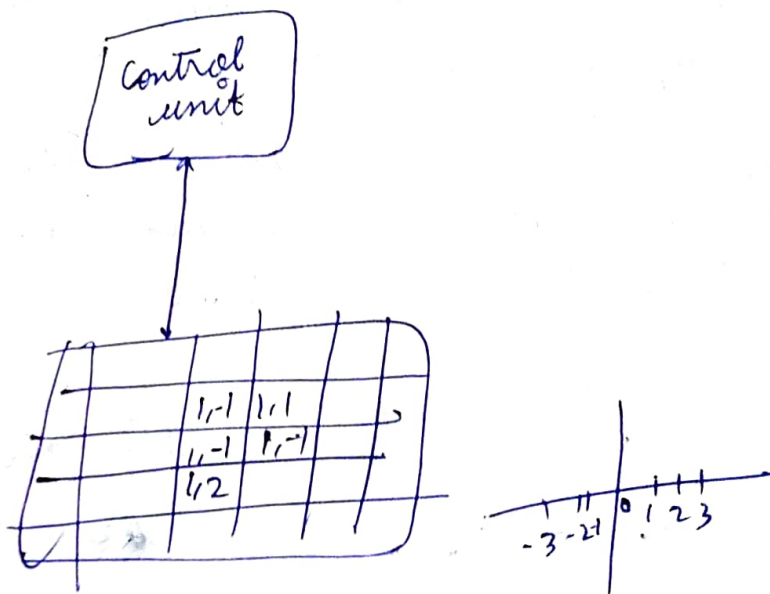
④ Offline TM:

- It has two types → one tape ^{is} reads only and contains the i/p.
→ other is read write and is blank initially.



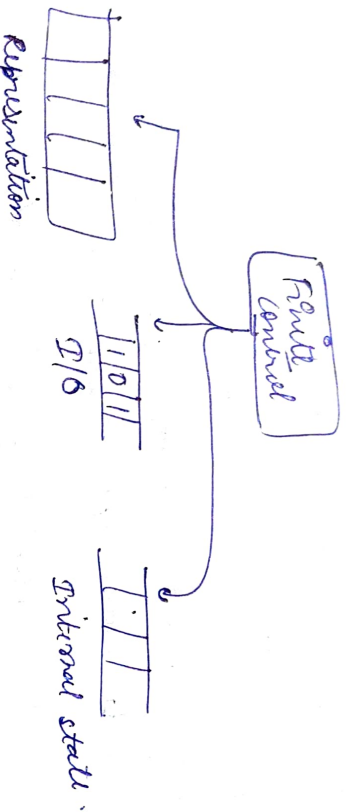
⑧ Multi Dimensional TM:

- It has multidimensional tape.
- e.g. A 2D TM will read/write on an infinite tape divided into ~~that~~ square like a checkboard.
- For a 2D TM transition function ~~stands~~ defined as -
$$\delta: Q \times T \rightarrow Q \times T \times (L, R, D).$$



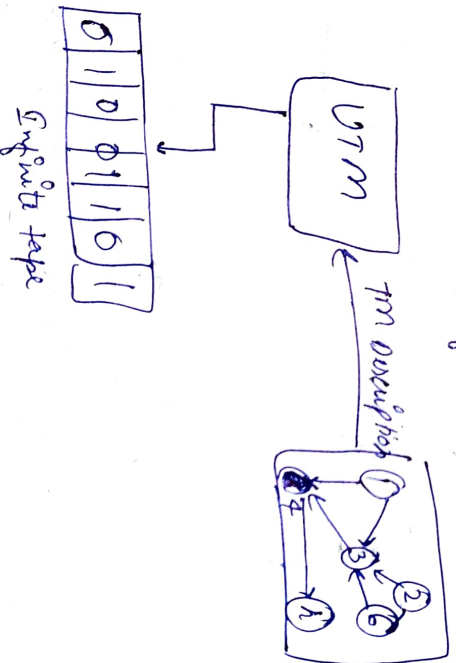
Universal TM

- The TM which behaves like a digital computer is called Universal TM. It is suggested by the mathematician Alan Turing during 1930. And has become the most widely used model of computation in computability and complexity theory.
- The model consists of input and output.
- The i/p is given in binary format form onto the tape and the o/p consists of the content of the tape when the machine halts.



of TM
 • The problem with the TM is that a different machine must be constructed for every new computation to be performed for every i/p & o/p relation.

• This is the reason the universal TM was introduced. A universal TM can simulate any other machine. The block diagram of universal TM is as follows -



• The universal TM includes 3 pieces of data for the machine it simulates:
 ↳ A basic description of the machine
 ↳ The content of the machine tape
 ↳ The internal state of the machine.